

Public Economics

Take-Home Assignment 1

Topics: Theoretical & Empirical Tools, Externalities

Name: _____ Student ID: _____ Due: _____

Answer both questions. Where a short explanation is requested, 2–3 sentences is enough — we are grading for correct reasoning, not length. Show your work for any calculation.

Question 1: Pigouvian Taxation

A café on Mariahilferstraße runs outdoor heaters in winter. Let Q be heater-hours per week. The café's private marginal cost of running the heaters is

$$PMC(Q) = 2Q,$$

and each heater-hour imposes a marginal external damage on the neighbors (noise, fumes) of

$$MD(Q) = Q.$$

The marginal benefit to café customers from heater-hours is

$$MB(Q) = 60 - Q.$$

- a) Write down the social marginal cost $SMC(Q)$. Find the socially efficient quantity of heater-hours, Q^* , and the Pigouvian tax per heater-hour that would achieve it.
- b) The district council is instead considering a flat rule: every café on the street may run heaters for at most 15 hours per week, no exceptions. In 2–3 sentences, explain why this uniform quantity cap could be *less* efficient than the Pigouvian tax from part (a), even if it happens to produce the same *total* heater-hours across the street.

Question 2: Designing a Natural Experiment

The city of Graz is banning through-traffic in its old town center to cut air pollution. The ban is **phased in** one neighborhood at a time over 18 months, rather than everywhere at once.

- a) A local newspaper wants to claim the ban “caused” lower pollution by simply comparing air-quality readings in the old town *before* vs. *after* the full ban took effect. In 2–3 sentences, explain why this before/after comparison alone is not credible evidence of a causal effect.
- b) The staggered rollout gives you a better option. Briefly describe a difference-in-differences design you could use instead: at any given point during the 18 months, which neighborhoods would be your *treatment* group and which would be your *control* group? What assumption about the control group does this design rely on?